



Explainable Artificial Intelligence for Plant Disease Detection: A Scoping Review of Models, Interpretability, and Real-World Challenges

Emmanuel Bulus Yahaya ¹, Imrana Yahaya Umar ², Maryam Abubakar Sharif ³, Aisha Muhammed Bose Ahmed ⁴, Usman Idris Ismail ^{5*}

¹⁻² Federal College of Horticulture, Dadin Kowa, Gombe State, Nigeria

³ Department of Computer Science, School of Science and Technology, Federal Polytechnic Kaltungo, Gombe State, Nigeria

⁴⁻⁵ Department of Computer Science, Federal University Kashere, Gombe State, Nigeria

* Corresponding Author: Usman Idris Ismail

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Abstract

The application of artificial intelligence (AI) in plant disease detection has rapidly advanced, enabling accurate, timely, and automated diagnosis from image data. Deep learning models, particularly convolutional neural networks (CNNs) and hybrid architectures, have demonstrated high classification performance, often exceeding 95% accuracy under controlled conditions. However, the lack of model interpretability remains a major barrier to real-world adoption in agricultural settings. To address this challenge, explainable artificial intelligence (XAI) techniques have been increasingly integrated to enhance transparency and user trust. This scoping review systematically examines recent studies (2021–2025) on AI-based plant disease detection with a focus on model architectures, explainability methods, datasets, and practical applicability. A total of 17 studies were analyzed to identify key trends and research gaps. The findings indicate that Grad-CAM, LIME, and SHAP are the most commonly used XAI techniques, providing visual and feature-level explanations that support model interpretability. Despite these advancements, significant challenges persist, including high computational cost, lack of standardized evaluation frameworks, and inconsistencies in explanation quality. Furthermore, the reliance on limited and curated datasets restricts model generalization under real-world conditions. The review highlights a critical gap between high-performing experimental models and their practical deployment in agriculture. Future research should focus on developing diverse datasets, lightweight and interpretable models, and standardized evaluation strategies to enable robust, transparent, and scalable AI systems for sustainable agriculture.

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1. Introduction

The application of deep learning (DL) in plant disease diagnosis has significantly transformed digital agriculture, enabling automated and accurate identification of foliar diseases from image data (Majdalawieh *et al.*, 2025) ^[20]. Advanced models such as convolutional neural networks (CNNs), transfer learning architectures, and hybrid frameworks have demonstrated exceptional performance in detecting plant diseases across a wide range of crops. Several recent studies report classification accuracies exceeding 95%, with some achieving near-perfect results under controlled conditions (Iman *et al.*, 2023; Gupta *et al.*, 2022; Chug *et al.*, 2022) ^[14, 10, 5]. These models are capable of capturing complex visual patterns such as lesion structure, color variation, and texture, which are often difficult to detect using traditional image processing techniques (Firouz & Sardari, 2022) ^[9]. Despite

these advancements, the practical adoption of such systems in real-world agriculture remains limited.

A key challenge hindering adoption is the lack of interpretability in deep learning models (Li *et al.*, 2022). These systems are often regarded as “black boxes,” where the decision-making process is not transparent to users. In agricultural contexts, where decisions directly impact crop health, pesticide use, and economic outcomes, the inability to explain model predictions reduces trust among farmers, agronomists, and plant pathologists (Emi-Johnson *et al.*, 2025) [6]. To address this limitation, Explainable Artificial Intelligence (XAI) has emerged as an essential research direction, aiming to make AI systems more transparent, interpretable, and trustworthy. Techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM), Local Interpretable Model-agnostic Explanations (LIME), and SHapley Additive exPlanations (SHAP) have been widely adopted to provide visual and feature-based explanations of model predictions (Kamolov, 2025; Ennab & Mccheick, 2025; S. M. Hussain *et al.*, 2022) [17, 7, 11]. These methods help validate whether models focus on biologically relevant regions, thereby improving confidence in AI-assisted decision-making.

Despite the growing integration of XAI in plant disease detection, existing research remains disjointed and lacks methodological consistency. Many studies apply explainability techniques without clearly justifying their selection or evaluating their effectiveness. Furthermore, most works emphasize technical implementation rather than biological interpretability, leading to a gap between AI development and practical agricultural needs. In addition, datasets used in these studies are often limited, crop-specific, and collected under controlled conditions, which restricts model generalization. Performance degradation has been observed in scenarios involving complex backgrounds, varying lighting conditions, and visually similar diseases (Senanayake, 2024; Pelosi *et al.*, 2025) [28, 27]. These limitations highlight the need for a more structured and comprehensive understanding of how AI and XAI methods are applied in plant disease diagnosis.

Based on these observations, two critical gaps can be identified. There is a lack of systematic synthesis of explainability methods in plant disease detection, making it difficult to determine which techniques are most effective under varying agricultural conditions and there is no unified framework linking model architectures, dataset characteristics, and interpretability approaches, which limits the reproducibility and scalability of existing solutions.

This scoping review aims to address these gaps by systematically mapping the current landscape of AI-based plant disease detection with a focus on explainability. The study seeks to analyze the types of models employed, the datasets and modalities used, the XAI techniques applied, and the key contributions and limitations of existing research. By providing a structured synthesis, this review aims to identify trends, challenges, and opportunities for future work.

The review is guided by the following research questions:

- What types of deep learning models and architectures are used in explainable plant disease detection?
- Which XAI techniques are applied, and how do they contribute to model interpretability?

- How are datasets and experimental settings influencing model performance and generalization?
- What are the key limitations and research gaps in current approaches?

This study contributes to the field in several ways. It provides a structured overview of AI models and XAI techniques used in plant disease detection, consolidating findings across multiple studies. It also examines the relationship between model design and interpretability outcomes, highlighting how methodological choices impact transparency and performance. Furthermore, it identifies key challenges, including dataset bias, lack of standardized evaluation frameworks, and limited real-world validation. By synthesizing these aspects, the review offers insights into developing robust, interpretable, and deployable AI systems for agriculture.

Recent studies have emphasized the growing importance of XAI in agricultural applications, particularly in addressing the transparency and trust issues associated with deep learning models. Existing reviews have explored aspects such as model performance, dataset challenges, and emerging imaging techniques. However, they often treat explainability as a secondary component. In contrast, this study integrates these disjointed perspectives into a unified synthesis that explicitly focuses on the interplay between model performance, interpretability, and real-world applicability.

The remainder of this paper is organized as follows. Section 2 describes the scoping review methodology, including the search strategy, eligibility criteria, and study selection process. Section 3 presents the results and analysis of the included studies, focusing on model architectures, XAI techniques, datasets, and performance trends. Section 4 discusses the key challenges, limitations, and future research directions. Finally, Section 5 concludes the paper by summarizing the main findings and their implications for the development of transparent and trustworthy AI systems in plant disease detection.

2. Methodology

2.1. Study Design

This study adopts a scoping review methodology to systematically map and synthesize the existing literature on artificial intelligence (AI)-based plant disease detection, with a particular focus on explainable artificial intelligence (XAI). A scoping review approach was chosen as it allows for a comprehensive exploration of a broad and evolving research domain, identifying key concepts, methodologies, and research gaps without restricting the analysis to narrowly defined outcomes. The review was conducted in accordance with the methodological framework proposed by the Joanna Briggs Institute and reported following the guidelines of the PRISMA-ScR (Tricco *et al.*, 2018) [29] figure 1 presents the flow of the methodology adopted for the inclusion criteria.

2.2. Search Strategy

A comprehensive literature search was conducted across multiple electronic databases, including IEEE Xplore, ScienceDirect, SpringerLink, and Google Scholar. The search covered studies published between 2021 and 2025 to capture recent advancements in the field.

A combination of keywords and Boolean operators was used to retrieve relevant studies. The search terms included: “*plant disease detection*,” “*deep learning*,” “*convolutional neural networks*,” “*explainable AI*,” “*XAI*,” “*Grad-CAM*,” “*LIME*,” “*SHAP*,” and “*precision agriculture*.” These terms were adapted as necessary for each database to ensure comprehensive coverage.

2.3. Eligibility Criteria

The inclusion and exclusion criteria were defined to ensure the relevance and quality of the selected studies.

Inclusion criteria:

- Studies focusing on AI or deep learning-based plant disease detection.
- Studies incorporating explainability techniques (e.g., Grad-CAM, LIME, SHAP).
- Peer-reviewed journal articles and conference papers.
- Studies published between 2021 and 2025.
- Articles written in English.

Exclusion criteria:

- Studies not related to plant disease detection.
- Articles without AI-based methods.
- Studies lacking explainability components.
- Review papers, editorials, and non-peer-reviewed articles.
- Studies with insufficient methodological details or results.

2.4. Study Selection Process

The study selection process followed a structured screening procedure based on the PRISMA framework. Initially, all retrieved records were imported into a reference management system, and duplicate entries were removed.

In the first stage, titles and abstracts were screened to exclude irrelevant studies. In the second stage, full-text articles were assessed against the eligibility criteria. Only studies that met all inclusion criteria were selected for final analysis. The selection process resulted in 17 relevant studies (S1-S17) included in this review as shown table 1 below with the summary of their research contributions.

2.5. Data Extraction

A standardized data extraction form was developed to systematically collect relevant information from each included study. The extracted data included:

- Study identifier (Study ID and reference),
- Model type and architecture,
- Dataset used,
- Imaging modality,
- Explainability techniques applied, and
- Key contributions and limitations.

The extracted information was organized into a summary table to facilitate comparison and synthesis across studies.

2.6. Data Analysis and Synthesis

A descriptive and thematic analysis approach was employed to synthesize the extracted data. The findings were categorized based on key aspects such as model architectures, XAI techniques, datasets, and application domains. Patterns, trends, and relationships across studies were identified and analyzed to provide a comprehensive overview of the research landscape.

The synthesis focused on evaluating model performance, interpretability, generalization capability, and real-world applicability. Additionally, common limitations and research gaps were identified to highlight areas requiring further investigation.

2.7. Quality Considerations

Although formal quality assessment is not mandatory in scoping reviews, methodological rigor was maintained by including only peer-reviewed studies and clearly defined selection criteria. Emphasis was placed on transparency, reproducibility, and comprehensive coverage of the literature. Potential biases, such as publication bias and dataset bias within individual studies, were acknowledged during analysis.

2.8. Limitations of the Methodology

This review has several limitations. First, the search was restricted to studies published in English between 2021 and 2025, which may have excluded relevant research outside this timeframe or in other languages. Second, the reliance on selected databases may have resulted in the omission of some studies. Third, as a scoping review, the study does not provide a quantitative meta-analysis or formal quality scoring of included papers. Despite these limitations, the methodology provides a structured and comprehensive overview of current research trends in AI-based plant disease detection with explainability.

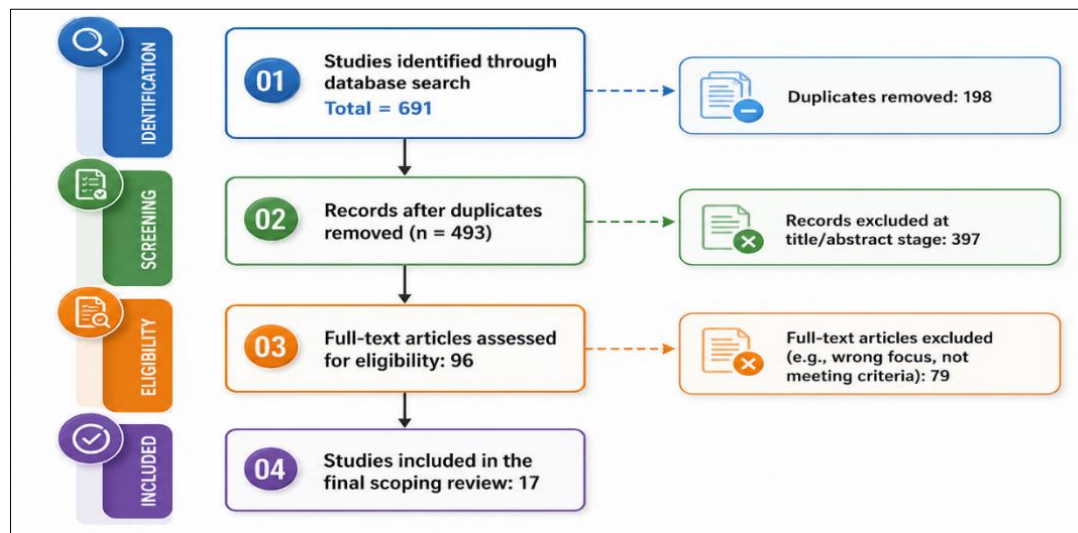


Fig 1: PRISMA Chart of The Studies

3. Results and Discussion

3.1. Overview of Included Studies

This scoping review analyzed 17 studies (S1-S17) published between 2021 and 2025, focusing on artificial intelligence-based plant disease detection with an emphasis on explainability. The findings reveal a strong trend toward the adoption of deep learning models, particularly convolutional neural networks (CNNs), for automated disease classification. Most studies reported high predictive performance, with accuracies typically exceeding 90% and in several cases approaching or reaching near-perfect classification (S1, S5, S9, S10, S16). These results

demonstrate the effectiveness of AI-based approaches under controlled experimental conditions.

In parallel, the integration of explainable artificial intelligence (XAI) techniques has become a central component of modern plant disease detection systems. Across the reviewed studies, methods such as Grad-CAM, LIME, SHAP, and Score-CAM were widely employed to improve model transparency and interpretability (S1, S6, S7, S14, S16). Despite these advancements, several limitations persist, particularly in terms of generalization, dataset diversity, and real-world deployment.

Table 1: Summary of the included studies

Study	Reference	Studies Contributions	Explainability used
S1	Preotee <i>et al.</i> , 2024 ^[23]	The study present VGG19 (98.90%) and Xception (98.66%) achieved the highest accuracies in leaf disease classification, though model generalization and real-world validation remain limited despite XAI integration.	GradCAM, GradCAM++, LayerCAM, ScoreCAM, FasterScoreCAM
S2	Partho <i>et al.</i> , 2024 ^[24]	The system achieved 99% accuracy using deep learning and IoT for real-time plant disease detection, with XAI highlighting infected areas. It enables low-cost smart farming but faces challenges like limited field testing, dataset bias, high computation, and overfitting risks.	Grad-CAM
S3	Khandaker <i>et al.</i> , 2024 ^[15]	ResNet50 achieved 90.5% accuracy, with CAM-based XAI methods improving interpretability. The study showed transparent pumpkin leaf disease detection but was limited by a small dataset and overfitting in deeper models.	Grad-CAM, Grad-CAM++, Score-CAM, Layer-CAM
S4	Vardhan <i>et al.</i> , 2024 ^[30]	The study shows that MobileNet outperformed other models with high accuracy and balanced metrics, while InceptionNet achieved perfect recall but showed misclassifications. The study faced challenges with hyperparameter sensitivity, limited dataset diversity, and high computational cost of Bayesian optimization.	LIME
S5	Peyal <i>et al.</i> , 2024 ^[25]	The proposed CNN-SVM model achieved 99.18% accuracy, outperforming VGG16 (93.42%) and VGG19 (91.94%) with a smaller, more efficient architecture. It offers high accuracy at low computational cost but is limited to grape leaf diseases and specific environmental conditions.	Grad-CAM
S6	Elsayad <i>et al.</i> , 2025 ^[8]	VGG19 achieved the highest accuracy (96.7%), followed by MobileNet-V2 (96.5%). XAI techniques enhanced model interpretability, but the dataset's four-disease scope limits real-world applicability, and high computational cost hinders mobile deployment.	LIME, Grad-CAM, Occlusion Sensitivity
S7	Peyal <i>et al.</i> , 2024 ^[26]	The CNN-SVM model achieved 98.34% accuracy, surpassing VGG16 and VGG19 while using only 0.95M parameters. XAI techniques improved model transparency, but the study was limited to four corn diseases and specific environmental conditions.	Grad-CAM, SHAP, LIME
S8	Maurmo <i>et al.</i> , 2024 ^[19]	The study achieved 92% overall accuracy and 98% for cassava mosaic disease using a transfer learning-XAI approach for transparent disease detection. Performance declined for visually similar diseases and under extreme lighting or mixed symptoms.	SHAP (SHapley Additive exPlanations)
S9	Islam <i>et al.</i> , 2023 ^[12]	The study achieved 100% accuracy across 39 crop diseases using a lightweight CNN with channel attention for efficient and precise classification. Grad-CAM enhanced	Grad-CAM

		interpretability, but data bias and absence of real-world validation remain limitations.	
S10	Otiya <i>et al.</i> , 2023 ^[22]	ResNet50 achieved the highest accuracy (99.1%) for cotton leaf disease detection, with Xception and DenseNet121 also performing well. Pre-trained models showed strong results, but dependence on specific datasets limits generalizability.	Grad-CAM
S11	Arvind <i>et al.</i> , 2021 ^[2]	EfficientNet-B5 achieved F1-scores of 0.9842 and 0.9930 for original and augmented datasets, respectively. The AI-based system enabled real-time, explainable plant disease detection with expert recommendations but struggled to classify multiple diseases per leaf, suggesting a multi-model approach for improvement.	LIME, Grad-CAM
S12	Noviandy <i>et al.</i> , 2023 ^[21]	The study showed that CLAHE enhanced CNN performance, improving accuracy and related metrics. Grad-CAM confirmed the model's focus on disease regions, demonstrating improved	Grad-CAM
S13	Ankatwar and Dhawale, 2024 ^[3]	DenseNet121 achieved 98.5% training and 97.85% testing accuracy, outperforming EfficientNet (94.9%). The study advanced crop disease detection using deep learning and XAI with spectral imaging but required high computation and large labeled datasets.	Not specified
S14	Kursun and Koklu, 2023 ^[16]	VGG16 achieved 98.88% accuracy by focusing on diseased regions, with Score-CAM providing visual explanations. The study improved explainability and early disease detection but was limited to a specific dataset and showed occasional inaccuracies in feature localization.	Score-CAM
S15	Islam <i>et al.</i> , 2024 ^[13]	The Dense Efficient model, combining DenseNet121 and EfficientNetV2B0, achieved near-perfect precision, recall, and F1-scores for watermelon disease detection. LIME and Grad-CAM improved interpretability, but the study lacked object annotation, tracking, and diverse datasets.	LIME, Grad-CAM
S16	Ahmed <i>et al.</i> , 2024 ^[1]	Exe-Net, an ensemble of Xception, DenseNet201, and InceptionResNetV2, achieved 99.75% accuracy with data augmentation. Grad-CAM, LIME, and SHAP enhanced explainability, though dataset imbalance and confusion between similar diseases like early and late blight remain challenges.	Grad-CAM, LIME, SHAP
S17	Bidouk <i>et al.</i> , 2023	The proposed XAI method outperformed D-RISE in detecting and localizing potato leaf diseases, offering model-agnostic explainability for any object detector. However, it requires high computational resources and is limited to potato leaf diseases.	Perturbation-based (D-RISE inspired)

3.2. Model Performance and Architectural Trends

The reviewed literature highlights the dominance of established CNN architectures, including VGG16, VGG19, ResNet50, DenseNet121, EfficientNet, and MobileNet. Among these, VGG19 consistently achieved high accuracy, reaching 98.90% in one study (S1) and 96.7% in another (S6). Similarly, ResNet50 demonstrated strong performance, achieving up to 99.1% accuracy in cotton leaf disease detection (S10) and 90.5% in pumpkin disease classification (S3). These results confirm the robustness of deep CNNs in extracting disease-relevant features.

Lightweight architectures such as MobileNet also showed promising results, outperforming other models in certain cases while maintaining computational efficiency (S4, S6). This makes them particularly suitable for real-time and resource-constrained applications. In addition, hybrid and ensemble approaches have further improved classification performance. CNN-SVM models achieved high accuracy with reduced computational complexity (S5, S7), while

ensemble frameworks such as Exe-Net achieved up to 99.75% accuracy by combining multiple deep learning models (S16). Preprocessing techniques and attention mechanisms also contributed to performance improvements. For example, CLAHE-based enhancement improved CNN accuracy by increasing feature visibility (S12), while channel attention mechanisms enabled precise feature selection, resulting in 100% accuracy in large-scale disease classification (S9). However, these high-performance results are often obtained using curated datasets, raising concerns about overfitting and limited real-world applicability. The distribution of deep learning models used across the reviewed studies has been presented in figure 2 below where CNN-based architectures, particularly VGG, ResNet, and DenseNet, are the most frequently adopted due to their strong performance in image classification tasks. The use of hybrid and ensemble models further reflects efforts to enhance predictive accuracy and robustness.

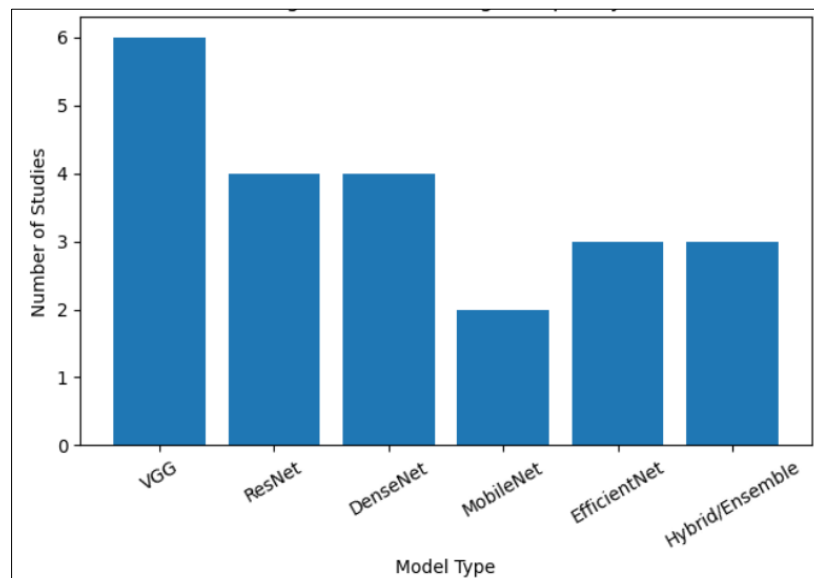


Fig 2: Model Usage Frequency

3.3. Explainability and XAI Integration

Explainability has emerged as a critical component in AI-based plant disease detection systems. Grad-CAM was the most widely used XAI technique, employed in the majority of studies to generate heatmaps highlighting disease-affected regions (S1, S2, S5, S9, S10, S12, S15, S16). These visual explanations helped validate that models were focusing on biologically meaningful areas, thereby enhancing user trust. Other XAI techniques provided complementary insights. LIME was used to generate local, interpretable explanations of model predictions (S4, S6, S11, S15, S16), while SHAP offered feature-level importance analysis, improving transparency in decision-making (S7, S8, S16). Score-CAM and Layer-CAM further improved localization of disease regions and explanation quality (S1, S3, S14). In one study, a perturbation-based method inspired by D-RISE demonstrated

improved object detection explainability compared to baseline approaches (S17). Despite these benefits, several challenges remain. XAI methods are often computationally intensive, limiting their feasibility for real-time deployment, particularly on mobile devices (S6, S17). Additionally, inconsistencies between different explanation methods can lead to ambiguity in interpretation. Most studies also rely on qualitative visual inspection of explanations rather than quantitative evaluation, indicating a lack of standardized metrics for assessing explainability effectiveness. Figure 3 below illustrates the proportion of explainability techniques employed in plant disease detection studies. Grad-CAM is the dominant method, widely used for generating visual explanations. Other techniques such as LIME and SHAP are also applied to provide complementary interpretability, highlighting the diversity of approaches in XAI integration.

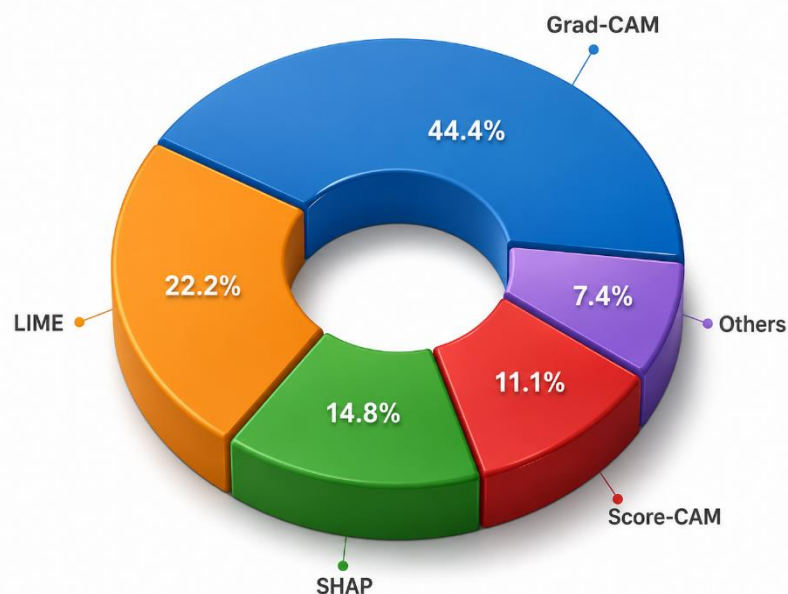


Fig 3: XAI Techniques Distribution

3.4. Datasets and Generalization Challenges

A key limitation identified across the reviewed studies is the reliance on limited and highly curated datasets. Many studies focused on specific crops or a small number of disease classes, such as grape (S5), corn (S7), pumpkin (S3), and watermelon (S15), which restricts the generalizability of the proposed models. Even studies with broader coverage reported potential dataset bias and lack of real-world validation (S1, S9, S10).

Performance degradation under real-world conditions was also highlighted. For instance, reduced accuracy was observed when distinguishing visually similar diseases or handling mixed symptoms (S8, S16). Environmental factors such as lighting variations and complex backgrounds further impacted model performance (S8). Additionally, some

studies reported overfitting issues due to small dataset sizes (S3), while others emphasized the need for more diverse and representative data (S6, S15). Although techniques such as data augmentation, preprocessing, and spectral imaging were used to mitigate these challenges (S12, S13), they do not fully address the fundamental issue of dataset bias. The lack of cross-dataset validation further limits confidence in the robustness of these models. The distribution of dataset types used in the reviewed studies is shown figure 4 below where most studies rely on controlled and single-crop datasets, while relatively few incorporate real-world data. This imbalance indicates a limitation in current research and highlights the need for more diverse datasets to improve model generalization.

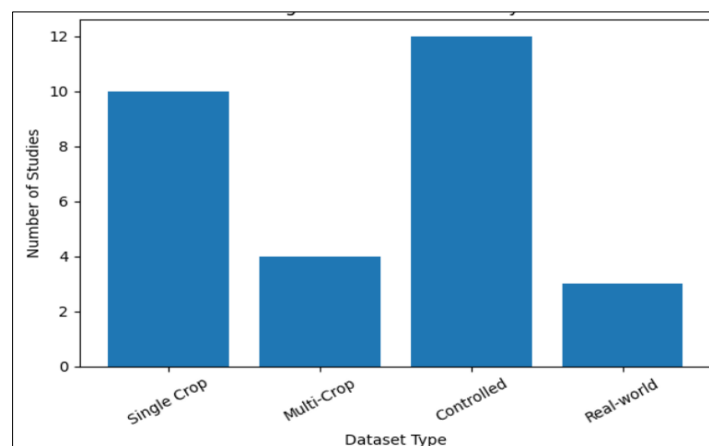


Fig 4: Dataset Diversity

3.5. Real-World Deployment and Practical Considerations

While most studies focused on model development and evaluation, only a few addressed real-world deployments. One notable approach involved integrating deep learning with IoT systems to enable real-time plant disease monitoring, supporting smart farming applications (S2). Additionally, lightweight models and efficient architectures demonstrated potential for deployment on mobile and edge devices (S4, S9).

However, practical implementation remains challenging. High computational requirements, particularly for deep learning and XAI techniques, hinder deployment in resource-constrained environments (S6, S17). Moreover, many models lack validation in real agricultural settings, where environmental variability and data noise can significantly affect performance. Another limitation is the lack of user-centered design. Although some systems provided explainable outputs and expert recommendations (S11), most studies focused primarily on classification accuracy without addressing usability or decision support. Features such as multi-disease detection, disease tracking, and actionable insights are often missing, limiting the practical value of these systems for farmers.

3.6. Evaluation Practices and Research Gaps

The evaluation of models across the reviewed studies primarily relied on metrics such as accuracy, precision, recall, and F1-score. While these metrics indicate strong predictive performance, they do not fully capture model robustness, interpretability quality, or real-world effectiveness. For

example, several studies reported near-perfect accuracy (S9, S16), yet acknowledged limitations related to dataset bias and lack of field validation.

A major gap in the literature is the absence of standardized evaluation frameworks. Few studies performed cross-dataset validation or assessed model performance under diverse environmental conditions. Similarly, the evaluation of XAI methods remains largely qualitative, with limited use of quantitative measures to assess explanation reliability.

These limitations highlight the need for more comprehensive evaluation approaches that consider multiple dimensions, including generalization, computational efficiency, and interpretability.

3.7. Discussion and Future Directions

The findings of this review reveal a clear gap between high-performing AI models developed in controlled environments and their real-world applicability. While deep learning models have achieved impressive accuracy, their ability to generalize to diverse agricultural conditions remains limited. The integration of XAI techniques has improved transparency and user trust, but challenges related to computational cost, consistency, and evaluation persist. Future research should focus on developing diverse and representative datasets that capture real-world variability, as well as designing lightweight models suitable for edge deployment. Establishing standardized benchmarks for both performance and explainability will be essential for meaningful comparison across studies. Additionally, incorporating domain knowledge and end-user feedback can enhance the usability and practical relevance of AI systems.

In summary, although the reviewed studies demonstrate significant progress in AI-based plant disease detection, achieving scalable and reliable real-world solutions requires addressing fundamental challenges in generalization, evaluation, and deployment.

4. Conclusion

This scoping review highlights the significant progress achieved in AI-based plant disease detection, particularly through the use of deep learning models and the integration of explainable artificial intelligence (XAI). The analyzed studies (S1–S17) demonstrate that models such as VGG, ResNet, DenseNet, and ensemble approaches can achieve high classification performance, often exceeding 95% accuracy (S1, S5, S10, S16). Additionally, the incorporation of XAI techniques, including Grad-CAM, LIME, and SHAP, has improved model transparency by providing visual and feature-based explanations that enhance user trust (S1, S6, S7, S16). Despite these advancements, several critical challenges remain. Most models are trained and evaluated on limited, curated datasets, raising concerns about their generalizability to real-world agricultural conditions (S3, S8, S15). Environmental variability, class imbalance, and visually similar disease patterns continue to affect model robustness. Furthermore, the computational complexity of both deep learning models and XAI methods limits their deployment on resource-constrained devices (S6, S17). Current evaluation practices also rely heavily on accuracy metrics, with limited emphasis on standardized benchmarking, cross-dataset validation, and quantitative assessment of explainability. In conclusion, while AI and XAI have shown strong potential for transforming plant disease detection, future research must focus on developing diverse datasets, efficient models, and standardized evaluation frameworks. Bridging the gap between high experimental performance and real-world applicability is essential for enabling reliable, scalable, and user-centered smart agriculture solutions.

Ethics statement Not applicable

This manuscript does not include human or animal research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI use

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to improve readability and language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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